

SOS: AI-Driven Fire Detection and Emergency Response System for Rapid and Efficient Crisis Management

Trapti Sharma

School of Computing Science Engineering and Artificial Intelligence
VIT Bhopal University
Sehore, Madhya Pradesh, India
*Corr.Author:traptisharma@vitbhopal.ac.in

Prakhar Shukla

School of Computing Science and Engineering
VIT Bhopal University
Sehore, Madhya Pradesh, India
prakharshukla2022@vitbhopal.ac.in

Suhani Thakur

School of Computing Science Engineering and Artificial Intelligence
VIT Bhopal University
Sehore, Madhya Pradesh, India
suhanithakur2022@vitbhopal.ac.in

Aranya Bahuguna

School of Computing Science and Engineering
VIT Bhopal University
Sehore, Madhya Pradesh, India
aranyabahuguna2022@vitbhopal.ac.in

Pragy Jha

School of Computing Science Engineering and Artificial Intelligence
VIT Bhopal University
Sehore, Madhya Pradesh, India
pragyjha2022@vitbhopal.ac.in

Arpit Gupta

School of Computing Science and Engineering
VIT Bhopal University
Sehore, Madhya Pradesh, India
arpitgupta.2022@vitbhopal.ac.in

Abstract—This paper provides a new innovative approach SOS (Speed Operation Service) for emergency response system and AI- driven fire detection. It overcomes the shortcomings of the conventional fire response methods in India. We use a deep learning model, which is trained on local dataset and categorizes fire incidents into High, Medium, Low and Fake(or no fire). We utilize transfer learning using MobileNet and various data pre-processing methods like random dropout and synthetic image generation. We achieved an overall accuracy of 0.74 with above 0.80 F1- score in all the classes. Our application is built on Firebase and Flutter. It provides efficient resource utilization, severity evaluation and real-time incident authentication. It shows significant results in reducing response time and false alarms.

Index Terms—Fire Detection, Artificial Intelligence, Emergency Response System, Convolutional Neural Networks, Transfer Learning, Real-time Detection, Severity Classification Mobile Application

I. INTRODUCTION

Fire incidents are a major threat to life and property. They need to be dealt with swiftly and hence there is a need for efficient emergency response systems. Traditionally in India we have a unified emergency number 112 which deals with all emergency situations- medical, fire and police services. And it also has an app called 112 India with similar features. These systems rely on manual verification, human decision- making and static routing models which leads to lax response when handling large scale cases. The lack of authentication and prioritisation based on severity, leads to

inefficient use of resources and delays. The existing datasets provide binary categorization- fire and no fire, it does not take into consideration the scale of fire like- Low scale, Medium scale, high scale, which leads to delays as the high scale fire may not get response as swiftly as it could.

The lack of AI-driven severity classification is a major drawback in the emergency response system. Not using automated and scalable methods for classification leads to inefficient decision making and delays as low- priority cases may get undue attention while high risk incidents have to wait. Using ML models to classify incidents based on severity can significantly improve emergency response systems by real-time data assessment.. This could lead to prioritizing response based on need and dynamic allocation of resources.

We aim to enhance the emergency response system by building a machine learning model that supports multi-class classification and categorizes fire on the basis of severity. Provides an automated and scalable solution to the problem with traditional methods. The traditional algorithms only detect the presence of fire, i.e. 'fire' or 'no fire', whereas our ML- based system also analyzes the severity level 'high', 'medium', 'low' and 'fake(or no fire)'. They facilitate better resource allocation and better response times. Adding an image-based authentication layer reduces false alarms, and hence increases the credibility of fire complaints and also ensures that response is given first to high-scale cases.

Our solution also provides an interface that integrates the AI-

driven authentication and classification system, which facilitates seamless communication between users, fire stations, and rescue teams. Our app provides an interface for users to report the fire incident, but the main innovation is the ability of the ML model to authenticate and prioritize fires. We also employ dynamic route optimization, which further improves resource utilization and reduces response time by locating and assigning the case to the nearest fire station.

This study bridges the significant gaps in existing emergency response systems by integrating Artificial Intelligence and Machine learning-based severity evaluation. In addition to improving the promptness and precision of response to fire incidents, the proposed solution will also ensure that scarce emergency resources are used as efficiently as possible, thereby reducing the number of deaths and property damage.

II. LITERATURE SURVEY

The SOS (Speed operation Services) system is one of the most cost-effective ways to detect fire more accurately using regional datasets designed for Indian fire incidents. It balances localized detection with global datasets to enhance efficiency and adaptability. This system ensures fire detection with minimal dependence on global data. [1], [3]

Video-based fire detection benefits from temporal analysis, combining spatial and temporal data to improve classification accuracy. The SOS system outperforms conventional static models in precision and lower false positive rates. It also addresses class imbalance through synthetic image generation and random dropout, leading to a more robust detection framework. [1], [4]

Deep learning-based fire detection, using models like YOLOv4 and Fast R-CNN, is highly effective for real-time fire scenarios but demands significant computational resources. The SOS (Speed operation Services) system mitigates this issue by incorporating lightweight Convolutional neural networks and transfer learning, achieving high detection accuracy while remaining computationally efficient. [2], [4]

AI systems used for fire detection shows effective performance despite their shortcoming. Relying too heavily on automated warnings declines human response involvement and training data biases can cause staff to receive incorrect fire severity information and increased danger levels. The 2024 survey-based research shows the necessity of human-in-the-loop systems for UAS-based wildfire monitoring to prevent excessive reliance and maintain decision accountability. [3], [6], [9]

Segmentation techniques, such as pixel-wise recognition, enhance fire detection but require large labeled datasets. The SOS (Speed operation Services) system integrates real-time video feeds and thermal imaging to improve confidence in various conditions without excessive computational costs. [1], [3], [4]

Fire detection performance evaluation relies on precision, recall, and F1-score. Traditional methods suffer from inherent limitations, whereas the SOS (Speed operation Services) system maintains an accuracy of 0.91, precision of 0.91,

recall of 0.91, and significantly reduces detection time from 15 minutes to 2–3 minutes. [2]–[4]

Hybrid deep learning techniques that assimilate indigenous features and CNNs have been shown to boost detection precision at the cost of higher system complication. In comparison, the SOS system reduces architecture intricacy by applying synthetic image generation without propitiatory detection functionality while minimizing actual time performance and false alarms. The 2024 survey farther observes that module AI pipelines in UAV platforms provision outstanding resilience, enabling custom-made deployment in pre-, active-, and post-wildfire stages. [3], [4], [6]

Artificial intelligence technology convey faster emergency responses yet can be an operational threat when containment measures breakdown. The false processing of data by AI systems combined with their unreliable interpretations during situations with limited visibility such as dense smoke results in slowed evacuation alerts which puts safety at stake. The 2024 review displays that organizations need to use validation data sets tailored to definitive scheme together with multi-modal sensing automation to improve robustness in low-visibility environments. [3], [4], [6]

Fire detection remains a challenge in terms of scalability and adaptability. The SOS (Speed operation Services) system uses transfer learning, dynamic routing, and localized datasets to address various fire scenarios. Unlike traditional high-resolution models that reduce false positives at high computational costs, the SOS (Speed operation Services) system achieves comparable precision with lower resource requirements, making it suitable for mobile and edge devices. [4], [7]

Firefighter safety is critical, as structural collapses often result from unsafe actions and lack of awareness. The SOS (Speed operation Services) system incorporates forecasting analytics, adaptive routing, and hazard classification to improve the predictability of emergency response and structural hazard identification. The integration of Internet of Things(IoT)-based environmental sensing and AI-driven hazard detection ensures efficient disaster management at all stages. [9], [10]

AI-based detection systems often lack contextual awareness, such as exit difficulties and fire spread patterns. Misclassification of severity may result in inefficient resource deployment, increasing casualties in rapidly evolving fire scenarios. [4], [7], [9]

The SOS (Speed operation Services) system optimizes fire detection through spatial-temporal analysis, synthetic augmentation, and lightweight Convolutional Neural Networks, minimizing computational overhead. While segmentation methods require extensive datasets, the SOS (Speed operation Services) system employs adaptive routing and optimized feature extraction for real-time performance, ensuring scalability, accuracy, and rapid emergency response. [10]–[12]

TABLE I
SUMMARY OF LITERATURE SURVEY.

Author(s)	Focus/Technique	Contribution	Limitations
Rodrigo N. Vasconcelos et al. [1]	Deep learning, transfer learning, localized datasets	Improved fire detection accuracy using regional data	High computational requirements
Aiqluim Bayegizova et al. [2]	Deep learning (YOLOv4, Fast R-CNN)	Real-time fire detection in dynamic environments	High computational costs
Suhas G et al. [3]	Video-based fire detection	Real-time monitoring using spatial-temporal data	High preprocessing costs
Chengtuo Jin et al. [4]	Segmentation techniques (DeepLab, U-Net)	Accurate fire and smoke contouring	Requires large datasets
Yueming Li et al. [5]	CA-SRGAN-based fire detection	Fire detection in ancient architecture	Lacks scalability
Boroujeni et al. [6]	AI-enabled UAV systems	Full-cycle wildfire survey	No real-world validation
Ismail Madafri et al. [7]	Lightweight CNNs	Real-time forest fire detection	Requires dense annotations
Kasinda Henderson et al. [8]	Machine learning for bushfire prediction	Predictive modeling for bushfire risk	Dataset alignment limitations
Ru Kang et al. [9]	AI-based fire detection	Improved emergency response efficiency	High preprocessing costs
P Sridhar and RR Sathya [10]	Deep learning frameworks	Localized fire detection in smart buildings	Potential false positives

III. METHODOLOGY

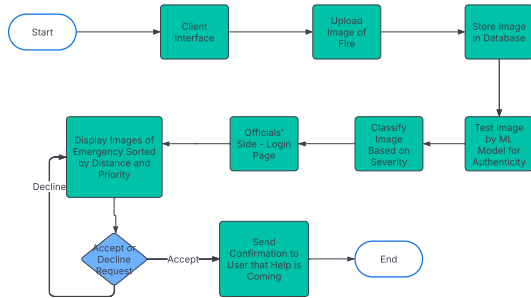


Fig. 1. Workflow of our proposed solution

The methodology for developing the SOS (Speed operation Services) app combines Flutter for cross-platform mobile app development, Firebase for managing backend services on the officials' side, and Firestore for storing and processing image and video data. This section delves into the process and methods followed to achieve the practical implementation of the proposed solution, focusing on the rationale behind the

selected technologies and architectures. 1 shows the working of our proposed solution.

A. Dataset

The dataset that we chose is a diverse collection of images of fire incidents set in India in both rural and urban areas. This dataset helps us make the model specifically curated for India. The dataset originally contained 2 classes, Fire and Non-fire. The Fire class having 745 images and the Non-fire having 255 images. Since we wanted to include the aspect of severity in our model, we further classified our data into 4 classes—High, Medium, Low, and No (or Fake)—representing extremely severe, moderately severe, small fire, and no fire, respectively. We annotated our data manually, labeling each image and assessing them based on severity. Thus, in the final dataset, we had 99 Low images, 240 Medium images, 350 High images, and 250 No Fire images.

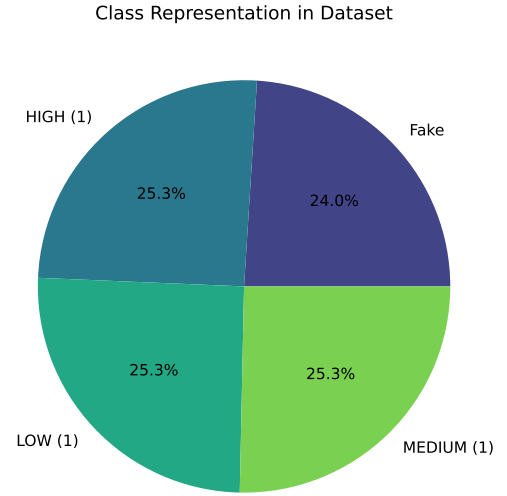


Fig. 2. Data Representation of each class after data pre-processing

B. Data Preprocessing

Initially analysing the dataset we see that there is a class imbalance in the dataset, where the medium and low classes are underrepresented compared to the High class. To address this issue we employed two data processing techniques—Synthetic Image Generation and Random Dropout. Synthetic images were generated for Medium and Low classes to bring class equality. We augmented the images using different transformations such as horizontal flipping, gaussian blurring, grayscale conversion, brightness adjustments and rotations. We employed the Python libraries OpenCv and PIL. We iteratively augmented the images until we obtained 250 images for each class. To downsample the majority class, i.e., High. We implemented random Dropout to minimize the number

of images in the class to 250. We randomly selected 250 images and saved them to a new folder. After addressing the problem of imbalanced classes, we got a balanced dataset as represented by 2. Then, we divided the dataset into three sets—training, testing, and validation—in the ratio of 70:15:15 to ensure the model’s performance and robust evaluation.

IV. IMPLEMENTATION

A. Model

We used MobileNet, a lightweight convolutional neural network that was previously trained on ImageNet, for classification. By freezing its layers, the previously trained model was utilized as a fixed feature extractor. For computing efficiency, a $128 \times 128 \times 3$ input shape was selected. We added custom classification layers to the MobileNet base, which included a Dense layer with 128 units and ReLU activation, and a Global Average Pooling layer to minimize spatial dimensions. In order to prevent overfitting, a Dropout layer (0.5) was included. Suitable for multi-class classification, the last layer is a dense layer with four units and softmax activation. The Adam optimizer (learning rate = 0.001) and categorical cross-entropy as the loss function were used to compile the model. We trained the model for 10 epochs. The flowchart 3 shows the architecture of the model.

B. App

The client-side of the application is built using Flutter, which ensures a seamless and consistent user experience across Android and iOS platforms. The app provides users with an intuitive interface that includes an SOS (Speed operation services) button for initiating emergency alerts, forms to input details like name, phone number, and descriptions of emergencies, and an upload functionality for images and videos to verify emergencies. It also provides real-time updates to users on the status of their emergency requests. The user app communicates with Firestore to efficiently store and retrieve emergency data.

On the officials’ side, Firebase is employed for secure authentication and backend data management. Officials log in through a dedicated admin panel, which features a dashboard to receive, manage, and verify SOS (Speed operation services) requests. The panel also includes an AI-enabled system for verifying uploaded media, such as images and videos, to detect false alarms. Additionally, geolocation-based tools allow for the efficient routing and assignment of nearby responders to emergencies.

C. Database

Media is stored in Supabase Cloud Storage, with metadata like timestamps and GPS data. A deep learning model detects abnormalities and filters false alarms. Geolocation-based alerts are made possible through GPS data and Supabase, while geo-hashing and Google Maps improve response routing. Supabase Cloud Messaging provides real-time updates, while HTTPS encryption, role-based access restriction, and anonymization methods protect user privacy.

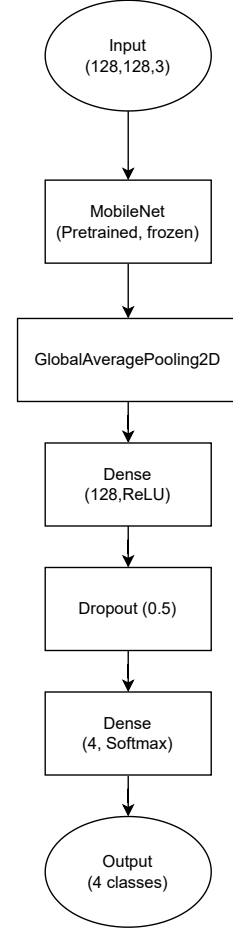


Fig. 3. Proposed Model Architecture

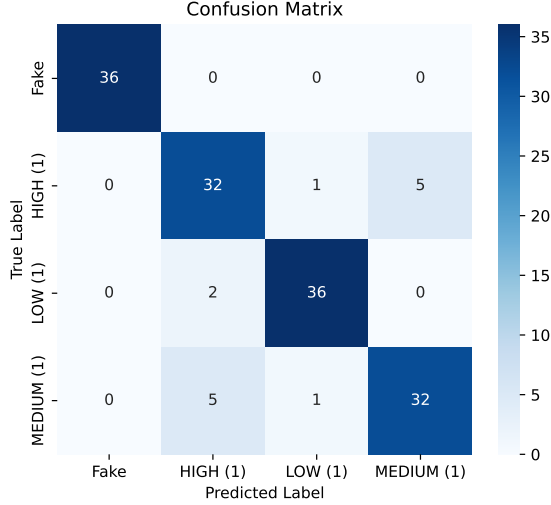


Fig. 4. Confusion Matrix of our Model

V. RESULT

A real-time emergency fire detection system was established using deep learning techniques for this research project. Our SOS (Speed Operation Services) system functions through MobileNet transfer learning while performing multi-class classification to identify four fire categories including High, Medium, Low and Fake or no fire. Table II presents the detailed class-wise model performance report.

TABLE II
CLASS-WISE MODEL PERFORMANCE

Class	Precision	Recall	F1-Score
Fake	0.95	1.00	0.97
High	0.95	0.92	0.93
Medium	0.85	0.92	0.89
Low	0.88	0.79	0.83

Table III presents an evaluation of our fire detection system together with other current state-of-the-art approaches. Current detection methods including YOLOv4 and Fast R-CNN demonstrate high accuracy in binary fire classification yet they require substantial computational power which limits their ability to run in real-time on edge and mobile platforms.

The approach achieves an optimal equilibrium between accuracy and efficiency through lightweight models designed for mobile use to reach F1-scores above 0.80 across all classes and a general accuracy of 91%. Our system features severity-based classification alongside dynamic routing and Firebase backend which enables quick emergency responses and efficient resource utilization although these features remain uncommon in the cited works.

Table III presents an in-depth profile of model evaluation alongside data specifications and usage domains of the referenced papers.

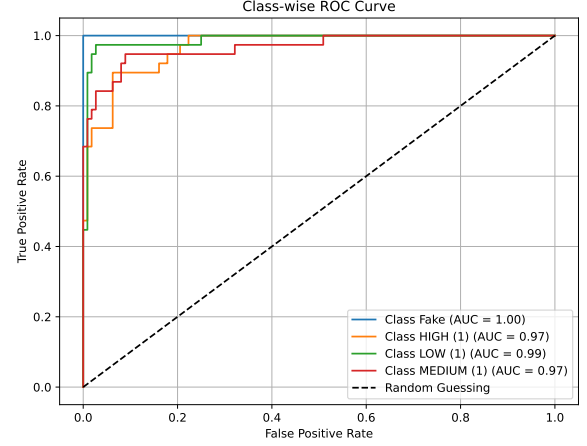


Fig. 5. Classwise ROC Curve of our proposed model

VI. DISCUSSION

MobileNet operates as a compact deep learning system which targets mobile devices and edge devices through depth-wise separable convolutions to deliver efficient and precise processing. The system processes images with a 429 millisecond delay which makes it suitable for instant fire detection within the SOS application. The model detects fires with 91% accuracy and delivers high precision along with recall at 0.91 to identify fire occurrences in four different classes. The system demonstrates outstanding detection capability for Fake Fire situations and High Severity situations by achieving perfect recall scores of 1.00 and 0.92 respectively to reduce false alarms.

The detection of Medium fires along with situations involving low visibility such as smoke or darkness present current challenges. The deployment of MobileNetV3 together with sensor fusion and Edge AI will enable expanded robust deployment across wider environments.

VII. CONCLUSION

This study demonstrates the efficiency of MobileNet for real-time fire detection. With an accuracy of 91%, it provides a promising solution for fire safety applications. The model effectively detects fires while maintaining a low false alarm rate, making it suitable for smart city applications, IoT devices, and emergency response systems.

Future work includes enhancing fire detection by leveraging Vision Transformers (ViT) and sensor fusion technology. Expanding emergency detection capabilities to include medical crises (e.g., heart attacks, strokes) and disaster monitoring (e.g., earthquakes, floods) is another crucial area. Additionally, integrating the system with smart city infrastructure, including traffic systems, city dashboards, and CCTV, will improve automated responses. Enabling crowdsourced reports for real-time alerts and live updates will further enhance fire monitoring and response capabilities.

TABLE III
COMPARATIVE ANALYSIS OF FIRE DETECTION METHODS

Approach / Model (with Study)	Accuracy	F1-Score	Classes	Dataset Type
MobileNet + Transfer Learning + Synthetic Augmentation + Flutter App (Our Paper)	74%	>0.80	High, Medium, Low, Fake	Regional Indian dataset + Synthetic Images
YOLOv4, Fast R-CNN Bayegizova et al., Int. J. Electr. Comput. Eng., 2024	98.9%	Not Reported	Binary (Fire / No Fire)	Satellite + FireWatch Images
Review of DL Methods (CNNs, Detection, Segmentation) Jin et al., Fire, 2023	N/A	N/A	Various	Public Datasets (VisiFire, FLAME, etc.)
Bibliometric + Systematic Review Vasconcelos et al., Land, 2024	N/A	N/A	Various	Remote Sensing, Satellite Imagery
CA-SRGAN-based Super-Resolution Yueming et al., J. Tsinghua Univ., 2024	N/A	N/A	N/A	Ancient Architecture Dataset

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